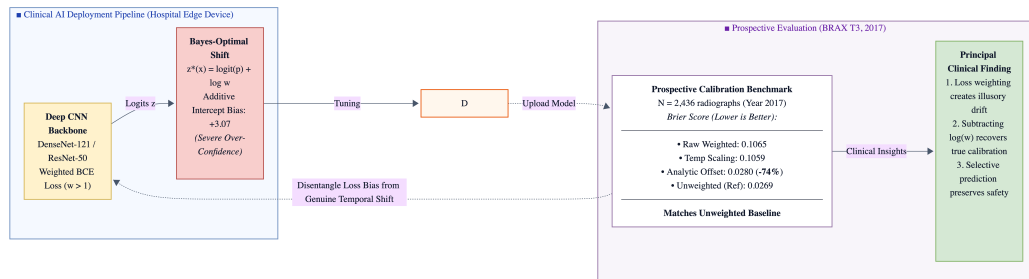


Graphical Abstract

Positive-Class Weighting Induces Systematic Probability Distortion in Deep Learning Models for Chest Radiography under Deidentified-Date Cohort Shift

Vishesh Panghal



Highlights

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- Positive loss weighting induces an additive population logit offset of $\log w$.
- Finite neural network logits empirically cluster tightly around theoretical $\log w$.
- Temperature scaling cannot correct this intercept-type calibration bias.
- Analytic offset ($z - \log w$) cuts 74% of Brier error without retraining.
- Selective triage exposes trade-offs between calibrated risk and screening referral.

Positive-Class Weighting Induces Systematic Probability Distortion in Deep Learning Models for Chest Radiography under Deidentified-Date Cohort Shift

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Abstract

Clinical deep learning models frequently exhibit calibration decay under deidentified-date cohort shift. Under extreme class imbalance, training with positive-weighted binary cross-entropy (BCE) is standard practice, but its downstream effect on probability calibration has been widely conflated with intrinsic feature drift. We resolve this conflation through a controlled $2 \times 2 \times 3$ factorial experiment across 12 models (DenseNet-121 and ResNet-50 trained under unweighted and positive-weighted BCE across 3 seeds) on 40,523 chest radiographs from BRAX. Models are trained on an anchor partition and evaluated across later deidentified-date strata (T_1, T_2, T_3) with strict patient isolation. We show that positive loss weighting mathematically induces an additive population logit shift of $\log w$, which temperature scaling structurally cannot correct. In finite networks, empirical logit differences cluster tightly around theoretical $\log w$ (mean $\Delta z = 3.29 \pm 0.81$ vs. theoretical 3.07 for Pleural Effusion). On stratum T_3 , zero-parameter analytic offset ($z - \log w$) reduces total Brier score by 74% ($\Delta \text{Brier} = -0.0795$ [95% CI: $-0.0893, -0.0697$]), substantially reducing intercept miscalibration from $\alpha = -2.96$ to -0.45 without retraining or validation data. On an external MIMIC-CXR-JPG cohort (3,403 radiographs, 289 patients), discrimination proved pathology-dependent (Pleural Effusion AUROC 0.688; Cardiomegaly 0.523), with severe baseline calibration shift ($\text{BSS} < 0$). Analytic offset yielded significant Brier reductions for Cardiomegaly and Pneumonia, but worsened

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Edema, delineating boundaries of zero-parameter recalibration under cross-institutional shift. Finally, selective prediction simulations expose operational trade-offs between risk estimation and high-sensitivity screening triage.

Keywords: Chest radiography, Probability calibration, Class imbalance, Distribution shift, Selective prediction, Deep learning, Clinical decision support

1. Introduction

Deep learning models for automated chest radiograph interpretation have achieved high discriminatory performance across common thoracic abnormalities such as pleural effusion, cardiomegaly, and pneumonia [1, 2]. However, the transition from stationary retrospective validation to longitudinal clinical deployment introduces substantial distribution shifts. In operational healthcare environments, patient populations evolve, imaging protocols shift, scanner hardware is replaced, and demographic or seasonal patterns alter condition prevalence over time [3, 4, 5].

While discriminatory ranking (measured via AUROC) often exhibits gradual longitudinal attenuation under temporal shift, decision-critical applications rely fundamentally on *calibrated posterior probabilities*: $P(Y = 1 \mid X = x)$ [6, 7]. When predicted probabilities deviate from true empirical event rates, downstream operational decision cutoffs shift unpredictably, expected utility formulations degrade, and automated decision-support pipelines generate distorted risk rankings.

Recent literature has reported rapid “temporal calibration decay,” asserting that frozen deep neural networks suffer progressive miscalibration over deployment periods [8, 9]. Simultaneously, because biomedical datasets are characterized by extreme class imbalance (pathology prevalence often below 5%), researchers routinely train vision models using positive-weighted binary cross-entropy (BCE) to penalize missed positive cases [1, 10].

In this work, we present a computational investigation showing that these two phenomena are deeply entangled, and that the prevailing attribution of temporal calibration degradation primarily to intrinsic feature drift may be largely explained by uncorrected post-training calibration offsets under label distribution shifts [11, 12]. Under population-risk minimization, when models are trained with positive loss weighting $w > 1$, the Bayes-optimal logit is shifted by an additive constant $+\log w$ [13]. Standard post-hoc

calibration methods—most notably temperature scaling [14]—only rescale the logit variance (z/T) and cannot shift the logit intercept. Consequently, weighted models continuously over-predict positive probabilities, producing severe Brier score and Expected Calibration Error (ECE) degradation that compounds over time as disease prevalence fluctuates.

1.1. Related Work

Probability Calibration in Deep Learning: Modern deep convolutional neural networks frequently exhibit over-confidence due to unconstrained cross-entropy minimization with over-parameterized architectures [14]. Post-hoc recalibration techniques such as temperature scaling, histogram binning, and isotonic regression have been widely evaluated [14, 15]. In medical decision models, calibration accuracy is paramount for valid risk stratification [6, 7]. However, standard methods were primarily derived for multi-class classification under balanced distributions, leaving binary imbalanced settings less formally explored.

Class Imbalance and Loss Function Design: Extreme class imbalance is pervasive in thoracic imaging. Common mitigation strategies include positive loss weighting [10, 1], focal loss [16], and re-sampling. While positive weighting successfully shifts operational sensitivity, King and Zeng [13] proved in classical logistic regression that sample proportion adjustments shift the model intercept by $\log w$. Modern deep learning pipelines routinely apply loss weighting without correcting this induced logit offset prior to post-hoc calibration.

Distribution Shift and Label Shift Adaptation: Temporal dataset shift encompasses changes in feature distributions $P(X)$, conditional distributions $P(Y | X)$, and prior label probabilities $P(Y)$ [3]. Davis et al. [12] demonstrated that calibration drift often precedes discriminatory loss in clinical models. Alexandari et al. [11] established that maximum likelihood with bias-corrected calibration provides robust adaptation under label shift. Decoupling algorithmic training offsets from genuine temporal shift is essential for diagnostic integrity.

Selective Prediction and Deferral Mechanisms: Selective classification allows models to abstain from predicting on ambiguous inputs, routing them to human experts [17, 18]. Fisch et al. [19] highlighted that selective classification performance is fundamentally bounded by probability calibration. Investigating how post-hoc calibration interacts with uncertainty-based triage under temporal shift addresses an open operational question.

67 *1.2. Contributions*

68 To disentangle algorithmic loss bias from distribution shift across cohort
69 strata, this paper provides:

- 70 1. **A Controlled Factorial Matrix:** We execute a $2 \times 2 \times 3$ factorial
71 design across 12 models (DenseNet-121 and ResNet-50 architectures $\times$
72 standard BCE and positive-weighted BCE $\times$ 3 random seeds) trained on
73 25,123 chest radiographs and evaluated on 8,302 held-out radiographs
74 across three sequential deidentified-date strata (T_1, T_2, T_3) from the
75 BRAX dataset [2].
- 76 2. **Theoretical Derivation and Empirical Verification of Logit
77 Shift:** Under population risk minimization, we derive that positive-
78 weighted BCE introduces a constant logit offset $\log w$. In finite neural
79 networks, we verify empirically that observed logit differences $(z_w - z_u)$
80 tightly mirror theoretical $\log w$ across multiple thoracic findings. We
81 show that temperature scaling cannot adjust the logit intercept, leaving
82 the calibration intercept severely biased ($\alpha \approx -2.96$), whereas an
83 exact zero-parameter analytic offset ($z - \log w$) or non-negative Platt
84 scaling restores calibration parity with unweighted models (substantially
85 reducing intercept bias to $\alpha = -0.45$ for DenseNet-121 and reducing
86 total Brier error by 74%).
- 87 3. **A Leak-Free Out-of-Sample Calibration Protocol:** We implement
88 a strict 5-fold patient-clustered cross-fitting protocol on T_1 , strictly
89 preventing in-sample calibration leakage, and report out-of-sample gen-
90 eralization on T_2 and T_3 backed by 2,000-replicate patient-clustered
91 bootstrap confidence intervals directly on individual seeds and the 3-seed
92 ensemble.
- 93 4. **Human-in-the-Loop Selective Prediction Simulation:** We for-
94 mulate a selective triage simulation with explicit, transparent tracking
95 of reader referral workloads, retained sensitivity $(TP/N_{\text{pos, ret}})$, and
96 full-cohort system sensitivity $(1 - FN/N_{\text{pos, total}})$ under matched de-
97 velopment thresholds, demonstrating that calibrated models optimize
98 probability estimation while raw weighted models support recall-oriented
99 screening triage.

100 2. Materials and Methods

101 2.1. Dataset and Cohort Stratification

102 We utilize the Brazilian Chest X-ray (BRAX) v1.1.0 benchmark [2]. BRAX
 103 v1.1.0 contains 40,967 radiographs. After excluding 444 radiographs from
 104 89 patients whose examinations crossed partition boundaries, the analytic
 105 cohort comprised 40,523 radiographs acquired across multiple inpatient and
 106 outpatient emergency facilities at Hospital Israelita Albert Einstein in São
 107 Paulo, Brazil. Radiographs are labeled with 14 thoracic findings extracted
 108 from Portuguese radiological reports using a validated natural language
 109 processing pipeline.

110 In compliance with HIPAA de-identification standards, direct patient
 111 identifiers and original timestamps are protected. We extract chronological
 112 study dates from the released 8-digit calendar integers (YYYYMMDD), spanning
 113 2008 through 2017. As is standard under HIPAA Safe Harbor de-identification,
 114 date anonymization may include patient-specific random date shifting that
 115 preserves relative within-patient temporal intervals while shifting absolute
 116 calendar offsets across patients. Consequently, our sequential partitions T_1
 117 (2015-H2), T_2 (2016), and T_3 (2017) are interpreted as chronologically ordered
 118 held-out deidentified-date strata under longitudinal distribution shift rather
 119 than synchronized real-world calendar years. To maintain strict partition
 120 integrity and preclude longitudinal data leakage:

- 121 1. All studies are partitioned chronologically around a strict model freeze
 122 boundary: $T_{\text{freeze}} = 2015-07-01$.
- 123 2. All training studies occur before 2013-09-01 ($N = 25,123$ images from
 124 11,265 patients across 14,315 studies).
- 125 3. All validation studies occur between 2013-09-01 and 2015-06-30 ($N =$
 126 7,098 images from 3,281 patients across 4,012 studies).
- 127 4. All held-out test studies occur between 2015-07-01 and 2017-12-17
 128 ($N = 8,302$ images from 3,807 patients across 4,670 studies), partitioned
 129 into three consecutive deidentified-date strata: T_1 (2015-H2, $N = 1,879$),
 130 T_2 (2016, $N = 3,987$), and T_3 (2017, $N = 2,436$).
- 131 5. Strict patient isolation is enforced: $\text{Patients}_{\text{train}} \cap \text{Patients}_{\text{val}} \cap \text{Patients}_{\text{test}} =$
 132 $\emptyset$. Specifically, 444 radiographs from 89 patients whose examinations
 133 crossed partition boundaries were excluded to guarantee zero within-
 134 patient contamination, yielding the analytic cohort of 40,523 radio-
 135 graphs.

Table 1: Cohort Demographics, Radiographic Projections, and Target Condition Prevalence across Stratified Cohorts

Cohort Stratum	Images (N)	Patients (N)	Studies (N)	Sex (% M / F)	Age (Mean $\pm$ SD)	Views (% PA / AP / Lat / Unk)	Pleural Effusion N (%)	Cardiomegaly N (%)	Pneumonia N (%)	Edema N (%)
Train (Pre-deployment)	25,123	11,265	14,315	49.3% / 50.7%	46.0 $\pm$ 24.3	34.0% / 13.5% / 0.0% / 27.2%	1,116 (4.44%)	2,398 (9.55%)	456 (1.82%)	23 (0.09%)
Val (Pre-deployment)	7,098	3,281	4,012	49.8% / 50.2%	46.2 $\pm$ 24.1	34.7% / 12.2% / 0.0% / 27.7%	295 (4.16%)	681 (9.59%)	128 (1.80%)	8 (0.11%)
Test T1 (2015)	1,879	913	1,039	47.8% / 52.2%	44.5 $\pm$ 23.9	34.4% / 10.2% / 0.0% / 29.0%	99 (5.27%)	147 (7.82%)	50 (2.66%)	4 (0.21%)
Test T2 (2016)	3,987	1,832	2,249	48.5% / 51.5%	46.1 $\pm$ 23.8	34.6% / 12.0% / 0.0% / 27.6%	177 (4.44%)	422 (10.58%)	79 (1.98%)	13 (0.33%)
Test T3 (2017)	2,436	1,164	1,382	50.9% / 49.1%	46.0 $\pm$ 24.6	34.4% / 12.4% / 0.0% / 28.1%	85 (3.49%)	258 (10.59%)	50 (2.05%)	0 (0.00%)
Test All (Pooled)	8,302	3,807	4,670	49.0% / 51.0%	45.7 $\pm$ 24.1	34.5% / 11.7% / 0.0% / 28.1%	361 (4.35%)	827 (9.96%)	179 (2.16%)	17 (0.20%)

6. For longitudinal paired logit offset and multi-calibrator evaluation, all 8,302 held-out radiographs ($N = 1,879$ in T_1 , $N = 3,987$ in T_2 , and $N = 2,436$ in T_3) are evaluated across both clinical architectures (DenseNet-121 and ResNet-50) with zero held-out exclusions (0% exclusion rate), yielding 16,604 paired model evaluations ($8,302 \times 2$) with complete longitudinal tracking.

Demographic, projection, and disease prevalence characteristics across all cohort strata are detailed in Table 1.

2.2. External Transport Validation Cohort (MIMIC-CXR-JPG)

To evaluate cross-institutional transportability under realistic clinical domain shift, frozen BRAX-trained models were externally evaluated on all eligible frontal radiographs (AP and PA views) from the official patient-partitioned MIMIC-CXR-JPG test partition [20, 21]. The cohort comprises $N = 3,403$ frontal radiographs ($N = 3,041$ diagnostic studies across $N = 289$ unique patients; AP projections: 2,405 [70.67%], PA projections: 998 [29.33%]). In strict accordance with the primary evaluation protocol, ground-truth diagnostic labels were extracted using the official CheXpert labeler under the standard uncertainty-zero (U -zero) policy. Empirical condition prevalence across images was: Pleural Effusion 32.18% ($N = 1,095$ positive radiographs), Cardiomegaly 26.33% ($N = 896$), Edema 21.33% ($N = 726$), and Pneumonia 10.05% ($N = 342$). Because multiple radiographs and studies belong to the same patient, patient-clustered non-parametric bootstrap inference was used throughout external evaluation to account for repeated within-patient observations, avoiding artificial inflation of effective sample size. All 3,403 radiographs were physically verified on disk (100.00% completeness). No retraining or hyperparameter fine-tuning was performed, isolating out-of-distribution transportability from in-domain adaptation.

2.3. Architectures and Training Objectives

We evaluate two standard clinical vision architectures initialized from ImageNet pre-trained weights:

- **DenseNet-121** [22]: Employs dense feature concatenation, standard for thoracic pathology benchmarks [1].
- **ResNet-50** [23]: Employs residual skip connections as an independent structural comparator.

Each architecture is trained under two distinct loss objectives:

1. **Unweighted Binary Cross-Entropy (Standard BCE):**

$$\mathcal{L}_{\text{std}}(y, \hat{p}) = -[y \log \hat{p} + (1 - y) \log(1 - \hat{p})] \quad (1)$$

where $\hat{p} = \sigma(z) = 1/(1 + e^{-z})$ is the sigmoid probability from raw logit z .

2. **Positive-Weighted Binary Cross-Entropy (Weighted BCE):**

$$\mathcal{L}_{\text{pos}}(y, \hat{p}) = -[w \cdot y \log \hat{p} + (1 - y) \log(1 - \hat{p})] \quad (2)$$

where $w = (N_{\text{total}} - N_{\text{pos}})/N_{\text{pos}}$ represents the inverse positive prevalence in the training set (e.g., $w = 21.51$ for Pleural Effusion, $w = 9.48$ for Cardiomegaly, $w = 54.09$ for Pneumonia).

Models are trained with the AdamW optimizer (initial learning rate 10^{-4} , weight decay 10^{-2}) and a Cosine Annealing learning rate schedule over 15 epochs, with batch size 64 and Automatic Mixed Precision (AMP fp16) on 512×512 pixel input resolution. Checkpoints are selected based on the epoch achieving the highest validation Pleural Effusion AUROC across the 15 epochs (occurring between epochs 4 and 15 across runs), without early stopping loop termination. Each configuration is executed across three random seeds ($S \in \{42, 123, 2026\}$), producing 12 fully independent models. For unified system-level benchmarking, a 3-seed probability ensemble is constructed for each architecture and loss objective by averaging predicted probabilities across independent training runs:

$$\bar{p}_{\text{raw}}(x) = \frac{1}{3} \sum_{s \in S} \sigma(z_s(x)), \quad \bar{p}_{\text{corrected}}(x) = \frac{1}{3} \sum_{s \in S} \sigma(z_s(x) - \log w) \quad (3)$$

All post-hoc calibration, paired analytic adjustments, and selective prediction policies are evaluated directly on both individual model seeds and the unified probability ensemble, ensuring strict mathematical consistency across all reported metrics and tables.

193 *2.4. Mathematical Mechanics of Loss Weighting and Calibration*

194 Under positive-weighted binary cross-entropy, the empirical risk minimizer
195 seeks to minimize the expected loss:

$$\begin{aligned}\mathbb{E}_{Y|X}[\mathcal{L}_{\text{pos}}(Y, \hat{p})] &= -w \cdot P(Y = 1 | X) \log \hat{p} \\ &\quad - P(Y = 0 | X) \log(1 - \hat{p})\end{aligned}\tag{4}$$

196 Setting the derivative with respect to logit z to zero:

$$\begin{aligned}\frac{\partial}{\partial z} \mathbb{E}[\mathcal{L}_{\text{pos}}] &= \hat{p} [wP(Y = 1|X) + P(Y = 0|X)] \\ &\quad - wP(Y = 1|X) = 0\end{aligned}\tag{5}$$

197 Solving for the optimal predicted probability $\hat{p}^*$:

$$\hat{p}^* = \frac{w \cdot P(Y = 1 | X)}{w \cdot P(Y = 1 | X) + P(Y = 0 | X)}\tag{6}$$

198 Converting to log-odds:

$$z^*(X) = \log \left(\frac{\hat{p}^*}{1 - \hat{p}^*} \right) = \log \left(\frac{P(Y = 1 | X)}{P(Y = 0 | X)} \right) + \log w\tag{7}$$

199 Equation (7) derives that positive loss weighting shifts the model’s population
200 Bayes-optimal logit by $+\log w$. When $w \gg 1$, the network is mathematically
201 incentivized to output large positive logits, systematically inflating predicted
202 probabilities. We emphasize that Equation (7) describes the population-risk
203 minimizer under infinite capacity; in finite neural networks trained with
204 stochastic gradient descent, empirical logits fluctuate around this theoretical
205 attractor.

206 *2.4.1. Limitation of Temperature Scaling*

207 Temperature scaling [14] adjusts probabilities via a single positive scalar
208 T :

$$p_{\text{temp}} = \sigma \left(\frac{z}{T} \right)\tag{8}$$

209 To formally evaluate calibration structure, we examine the standard logistic
210 calibration regression (Cox calibration) [6, 7]:

$$\text{logit}(P(Y = 1 | X)) = \alpha + \beta \cdot \text{logit}(\hat{p})\tag{9}$$

where $\alpha = 0$ represents perfect calibration-in-the-large (calibration intercept), and $\beta = 1$ denotes ideal calibration slope. Because positive loss weighting induces an additive shift $z \approx z_{\text{true}} + \log w$, the uncorrected logit suffers a massive intercept discrepancy ($\alpha \approx -\log w \ll 0$). Temperature scaling multiplies the logit by $1/T$, adjusting the slope $\beta \rightarrow \beta \cdot T$, but is mathematically incapable of shifting the additive constant α . In low-prevalence clinical domains where negative cases vastly dominate, the negative log-likelihood optimizer is constrained by the conflicting pressures of slope and intercept, typically selecting $T^* \approx 1.0$ or providing limited, architecture-dependent variance shrinkage without eliminating the severe underlying intercept displacement.

2.4.2. Post-Hoc Recalibration Suite

To systematically rectify this mechanism, we evaluate four post-hoc recalibration functions:

1. **Temperature Scaling (T^*):** $p = \sigma(z/T^*)$, optimized via bounded Brent minimization on $[0.01, 50.0]$ with an explicit monotonicity constraint (if $\text{NLL}(T^*) > \text{NLL}(1.0) \implies T^* = 1.0$).
2. **Analytic Offset ($z - \log w$):** Directly subtracts the theoretical training logit shift:

$$p_{\text{analytic}} = \sigma(z - \log w) \quad (10)$$

Requires zero validation data and zero numerical optimization, resetting $\alpha \approx 0$ by construction while preserving model discrimination and slope.

3. **Fitted Offset ($z + a^*$):** Optimizes a single scalar intercept $a^* \in [-10.0, 10.0]$ via univariate Brent minimization on validation log-likelihood, holding the slope fixed at 1.0.
4. **Platt Scaling ($a + bz$):** Logistic regression on logits $p = \sigma(a + bz)$ with a non-negative constraint on slope ($b \geq 0$) to prevent spurious rank reversal.

2.4.3. Algorithmic and Computational Complexity

The evaluated post-hoc calibrators exhibit distinct computational profiles:

- **Temperature Scaling:** Incurs $O(N_{\text{val}})$ scalar optimization over validation logits and $O(1)$ inference per prediction.
- **Analytic Offset:** Incurs strictly $O(1)$ closed-form scalar subtraction ($z - \log w$), requiring zero validation storage, zero parameter fitting, and $O(1)$ inference.

- 244 • **Fitted Offset:** Incurs $O(N_{\text{val}})$ univariate Brent optimization on valida-
245 tion log-likelihood and $O(1)$ inference.
- 246 • **Platt Scaling:** Incurs $O(k \cdot N_{\text{val}})$ iteratively reweighted least squares
247 fitting (where k is the number of optimization steps) and $O(1)$ inference.

248 The analytic offset thus provides an operational advantage for zero-compute
249 edge deployment, eliminating the requirement to retain or access historical
250 validation datasets.

251 2.5. Cross-Fitting and Resampling Evaluation Protocol

252 To ensure zero optimistic in-sample bias:

- 253 • On deployment stratum T_1 (2015-H2), calibrator parameters are esti-
254 mated using **5-fold patient-clustered cross-fitting**. Calibrators are
255 fit on 4 folds and evaluated out-of-fold on the 5th fold, with all studies
256 from any given patient assigned to a single fold.
- 257 • For evaluation on later held-out deidentified-date strata T_2 (2016) and
258 T_3 (2017), the calibrators fit on T_1 are frozen and applied strictly
259 out-of-sample.
- 260 • **Statistical Inference:** All 95% confidence intervals (CIs) are com-
261 puted using $B = 2,000$ patient-clustered bootstrap resamples with
262 replacement. For paired calibrator comparisons (e.g., Analytic Offset
263 vs. Raw), differences are computed on identical bootstrap resamples.

264 2.6. Selective Prediction and Deferral Policy

265 In operational deployment, models need not provide an autonomous pre-
266 diction for every image. An uncertain model can selectively defer ambiguous
267 examinations to expert human readers [17, 18].

268 To ensure fair comparisons across calibrators operating on different proba-
269 bility scales, we determine an operating decision cutoff t^* for each method
270 strictly on development stratum T_1 using Youden’s J index ($J = \text{Sensitivity} +$
271 $\text{Specificity} - 1$), without using later held-out evaluation labels. For unweighted
272 models, $t^* \approx 0.014$; for analytic offset calibrated models, $t^* \approx 0.022$; and for
273 raw weighted models, $t^* \approx 0.33$. We define a threshold-relative uncertainty
274 metric based on piecewise distance from the operational cutoff t^* :

$$u(x) = 1 - \frac{|\hat{p}(x) - t^*|}{\mathbb{I}(\hat{p}(x) \geq t^*)(1 - t^*) + \mathbb{I}(\hat{p}(x) < t^*)t^*} \in [0, 1] \quad (11)$$

where $u(x) \rightarrow 1$ indicates maximal ambiguity ($\hat{p}(x) \approx t^*$) and $u(x) \rightarrow 0$ indicates decisive confidence. Crucially, because t^* scales with baseline disease prevalence (e.g., $t^* \approx 0.022$ for analytic calibrated models), predictions below t^* are normalized by t^* . Consequently, low-probability false negatives ($\hat{p} \ll t^*$) yield low uncertainty and are retained in the automated pool rather than referred. At coverage level $C \in [0.5, 1.0]$, the automated system retains the fraction C of cases with lowest uncertainty ($N_{\text{auto}} = \lfloor C \cdot N \rfloor$) and refers the remaining fraction $1 - C$ ($N_{\text{ref}} = N - N_{\text{auto}}$) to radiologist review.

To prevent metric conflation and provide complete arithmetic transparency, we evaluate triage outcomes using distinct, explicit definitions:

- **Retained Positives ($N_{\text{pos, ret}}$):** The number of true positive cases remaining in the automated cohort after ambiguous examinations have been referred to human readers.
- **Automated True Positives (TP_{auto}):** True pathology cases correctly flagged as positive by the autonomous system ($TP_{\text{auto}} = \sum_{i \in \text{Auto}} \mathbb{I}(y_i = 1 \wedge \hat{y}_i = 1)$).
- **Automated False Negatives (FN_{auto} , Missed Cases):** True pathology cases erroneously classified as negative by the autonomous system ($FN_{\text{auto}} = \sum_{i \in \text{Auto}} \mathbb{I}(y_i = 1 \wedge \hat{y}_i = 0)$). Note that $TP_{\text{auto}} + FN_{\text{auto}} = N_{\text{pos, ret}}$.
- **Retained Sensitivity:** Diagnostic sensitivity evaluated strictly among the retained automated decisions: $\text{Sens}_{\text{retained}} = \frac{TP_{\text{auto}}}{N_{\text{pos, ret}}} = \frac{TP_{\text{auto}}}{TP_{\text{auto}} + FN_{\text{auto}}}$.
- **Full-Cohort Triage Sensitivity:** System-wide detection sensitivity across all $N_{\text{pos, total}}$ cases in the evaluation stratum (e.g., $N_{\text{pos, total}} = 85$ in T_3), assuming that all examinations deferred to human readers receive expert diagnostic review and are correctly diagnosed: $\text{Sens}_{\text{triage}} = \frac{TP_{\text{auto}} + N_{\text{pos, ref}}}{N_{\text{pos, total}}} = 1 - \frac{FN_{\text{auto}}}{N_{\text{pos, total}}}$.
- **Referral Workload:** The fraction and count of examinations routed to readers ($N_{\text{ref}} = (1 - C) \cdot N$).

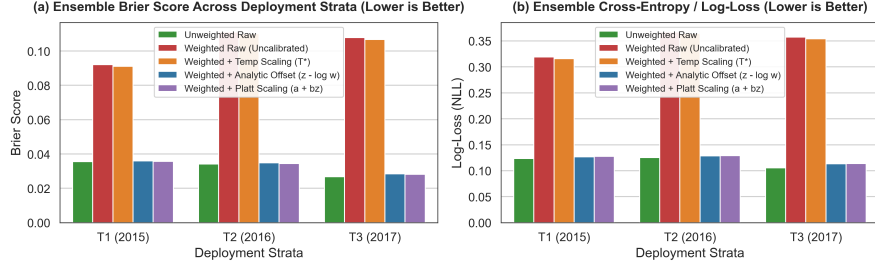


Figure 1: Multi-calibrator evaluation on Pleural Effusion for the DenseNet-121 3-seed ensemble across sequential deidentified-date strata (T_1, T_2, T_3), directly visualizing Table 3. (a) Brier score and (b) Log-loss. Temperature scaling fails to reduce probability error in weighted models, whereas Analytic Offset ($z - \log w$) and Platt Scaling eliminate 74% of Brier error, matching unweighted models.

3. Experimental Results

3.1. Factorial Matrix: Discrimination vs. Calibration Trajectories

Table 2 presents the comprehensive evaluation of the $2 \times 2 \times 3$ factorial matrix across DenseNet-121 and ResNet-50 for Pleural Effusion, Cardiomegaly, and Pneumonia across chronological strata T_1, T_2, T_3 .

3.1.1. Discrimination Attenuation

Both architectures exhibit moderate attenuation in AUROC across subsequent deidentified-date cohort strata. For DenseNet-121 on Pleural Effusion, AUROC decreases from 0.9335 ± 0.0052 at T_1 to 0.8731 ± 0.0120 at T_3 ($\Delta\text{AUROC} = -0.0604$). ResNet-50 exhibits an identical downward trajectory ($0.9295 \pm 0.0113 \rightarrow 0.8608 \pm 0.0190$, $\Delta\text{AUROC} = -0.0687$). This confirms that moderate discriminatory attenuation occurs across subsequent cohort strata, regardless of the training loss objective.

3.1.2. The Calibration Divergence and Factorial Interaction

In stark contrast to discrimination, probability calibration diverges completely depending on the training loss function:

- **Unweighted Models:** Brier scores are exceptionally low and actually *improve* across subsequent strata for conditions with falling prevalence. For unweighted DenseNet-121 on Pleural Effusion, the Brier score improves from 0.0369 ± 0.0029 (T_1) to 0.0279 ± 0.0009 (T_3), while Equal-Mass ECE decreases from 0.0232 to 0.0090.

325 • **Positive-Weighted Models:** Brier scores are dramatically elevated
 326 and appear to degrade across subsequent cohort strata ($0.1081 \pm$
 327 $0.0394 \rightarrow 0.1254 \pm 0.0456$ for DenseNet-121; $0.1256 \pm 0.0325 \rightarrow 0.1392 \pm$
 328 0.0310 for ResNet-50). ECE exceeds 22% across all held-out deidentified-
 329 date strata.

330 To formally examine the interaction between loss weighting and temporal
 331 cohort (LossWeighting $\times$ TemporalCohort), we evaluate the excess calibration
 332 error $\Delta\text{Brier} = \text{Brier}_{\text{weighted}} - \text{Brier}_{\text{unweighted}}$ across sequential strata:

333 • **Pleural Effusion (Prevalence Declines: 5.27% $\rightarrow$ 4.44% $\rightarrow$ 3.49%):**
 334 The excess calibration penalty of positive weighting expands mono-
 335 tonically across cohorts, from $\Delta\text{Brier} = +0.0712 \pm 0.0413$ in T_1 to
 336 $+0.0932 \pm 0.0428$ in T_2 and $+0.0975 \pm 0.0454$ in T_3 .

337 • **Cardiomegaly (Prevalence Increases: 7.82% $\rightarrow$ 10.58% $\rightarrow$ 10.59%):**
 338 The excess calibration penalty remains stable or slightly attenuates:
 339 $\Delta\text{Brier} = +0.0729 \pm 0.0255$ in T_1 , $+0.0681 \pm 0.0329$ in T_2 , and $+0.0683 \pm$
 340 0.0320 in T_3 .

341 • **Pneumonia (Prevalence Steady Low: 2.66% $\rightarrow$ 1.98% $\rightarrow$ 2.05%):**
 342 The penalty remains consistently elevated: $\Delta\text{Brier} = +0.1079 \pm 0.0526$
 343 in T_1 , $+0.1126 \pm 0.0552$ in T_2 , and $+0.1112 \pm 0.0508$ in T_3 .

344 This establishes the operational mechanism: a static training-induced positive
 345 logit offset produces surplus false-positive probability mass that is penalized
 346 more severely when deployment disease prevalence drops, expanding the
 347 apparent temporal calibration gap.

348 3.2. Multi-Calibrator Comparison

349 Table 3 and Fig. 1 present the multi-calibrator benchmark for Pleural
 350 Effusion on the 3-seed ensemble across 2,000 patient-clustered bootstrap
 351 resamples, including Cox calibration regression intercept (α) and slope (β)
 352 parameters.

353 3.2.1. Temperature Scaling Analysis

354 For weighted DenseNet-121, temperature scaling optimizes $T^* \approx 1.0$,
 355 producing an uncalibrated T_3 Brier score of 0.1067 compared to raw 0.1079.
 356 The paired bootstrap contrast on the ensemble is $\Delta\text{Brier} = -0.0011$ [95%

Table 2: Factorial Matrix Evaluation: Primary Longitudinal Performance across Architectures, Loss Objectives, and Thoracic Pathologies (Strata $T_1 \rightarrow T_3$). Values represent mean $\pm$ standard deviation across three independently trained random seeds ($S \in \{42, 123, 2026\}$). Complete 17-column factorial trajectories across all intermediate deployment strata (T_1, T_2, T_3) and log-loss metrics are detailed in Supplementary Table S3.

Architecture	Objective	Pathology	AUROC ($T_1 \rightarrow T_3$)	Δ AUROC	Brier ($T_1 \rightarrow T_3$)	Δ Brier	ECE T_3
DenseNet-121	Unweighted BCE	Pleural Effusion	0.934 $\rightarrow$ 0.873	-0.060	0.037 $\rightarrow$ 0.028	-0.009	0.0090
DenseNet-121	Unweighted BCE	Cardiomegaly	0.908 $\rightarrow$ 0.902	-0.006	0.053 $\rightarrow$ 0.068	+0.015	0.0210
DenseNet-121	Unweighted BCE	Pneumonia	0.847 $\rightarrow$ 0.865	+0.017	0.024 $\rightarrow$ 0.019	-0.005	0.0095
DenseNet-121	Unweighted BCE	Edema*	0.540 $\rightarrow$ -	-	0.002 $\rightarrow$ 0.000	-0.002	0.0029
DenseNet-121	Pos-Weighted BCE	Pleural Effusion	0.906 $\rightarrow$ 0.839	-0.067	0.108 $\rightarrow$ 0.125	+0.017	0.2227
DenseNet-121	Pos-Weighted BCE	Cardiomegaly	0.874 $\rightarrow$ 0.858	-0.016	0.126 $\rightarrow$ 0.136	+0.010	0.1969
DenseNet-121	Pos-Weighted BCE	Pneumonia	0.818 $\rightarrow$ 0.842	+0.024	0.131 $\rightarrow$ 0.130	-0.002	0.2425
DenseNet-121	Pos-Weighted BCE	Edema*	0.612 $\rightarrow$ -	-	0.061 $\rightarrow$ 0.064	+0.002	0.1274
ResNet-50	Unweighted BCE	Pleural Effusion	0.930 $\rightarrow$ 0.861	-0.069	0.037 $\rightarrow$ 0.028	-0.009	0.0132
ResNet-50	Unweighted BCE	Cardiomegaly	0.890 $\rightarrow$ 0.879	-0.011	0.057 $\rightarrow$ 0.071	+0.014	0.0175
ResNet-50	Unweighted BCE	Pneumonia	0.843 $\rightarrow$ 0.880	+0.037	0.024 $\rightarrow$ 0.019	-0.005	0.0106
ResNet-50	Unweighted BCE	Edema*	0.556 $\rightarrow$ -	-	0.002 $\rightarrow$ 0.000	-0.002	0.0017
ResNet-50	Pos-Weighted BCE	Pleural Effusion	0.879 $\rightarrow$ 0.819	-0.060	0.126 $\rightarrow$ 0.139	+0.014	0.2751
ResNet-50	Pos-Weighted BCE	Cardiomegaly	0.830 $\rightarrow$ 0.799	-0.031	0.142 $\rightarrow$ 0.152	+0.011	0.2338
ResNet-50	Pos-Weighted BCE	Pneumonia	0.835 $\rightarrow$ 0.823	-0.012	0.167 $\rightarrow$ 0.169	+0.002	0.3253
ResNet-50	Pos-Weighted BCE	Edema*	0.549 $\rightarrow$ -	-	0.069 $\rightarrow$ 0.075	+0.005	0.1670

Table 3: Multi-Calibrator Comparison on Pleural Effusion: Deidentified-Date Ensemble Performance, Calibration Intercept (α), Slope (β), and Paired Bootstrap Contrasts. Complete 12-column factorial evaluation across all intermediate strata and log-loss is provided in Supplementary Table S4.

Architecture	Loss Objective	Calibrator	Brier ($T_1 \rightarrow T_3$)	T_3 α	T_3 β	T_3 ECE	T_3 Δ Brier vs Raw [95% CI]
DenseNet-121	Weighted BCE	Raw (Uncalibrated)	0.092 $\rightarrow$ 0.108	-2.964	1.048	0.2227	-
DenseNet-121	Weighted BCE	Temperature Scaling (T^*)	0.091 $\rightarrow$ 0.107	-2.962	1.029	0.2190	-0.0011 [-0.0014, -0.0009]
DenseNet-121	Weighted BCE	Analytic Offset ($z - \log w$)	0.036 $\rightarrow$ 0.028	-0.445	0.929	0.0126	-0.0795 [-0.0893, -0.0697]
DenseNet-121	Weighted BCE	Platt Scaling ($a + bz$)	0.036 $\rightarrow$ 0.028	-0.229	1.128	0.0176	-0.0798 [-0.0894, -0.0702]
DenseNet-121	Unweighted BCE	Raw Baseline	0.036 $\rightarrow$ 0.027	+0.147	1.044	0.0040	-
ResNet-50	Weighted BCE	Raw (Uncalibrated)	0.110 $\rightarrow$ 0.123	-2.853	1.950	0.2751	-
ResNet-50	Weighted BCE	Temperature Scaling (T^*)	0.088 $\rightarrow$ 0.101	-2.608	1.184	0.1993	-0.0220 [-0.0241, -0.0198]
ResNet-50	Weighted BCE	Analytic Offset ($z - \log w$)	0.040 $\rightarrow$ 0.029	+1.075	1.405	0.0158	-0.0936 [-0.1022, -0.0844]
ResNet-50	Weighted BCE	Platt Scaling ($a + bz$)	0.038 $\rightarrow$ 0.030	-0.191	1.178	0.0223	-0.0930 [-0.1007, -0.0849]
ResNet-50	Unweighted BCE	Raw Baseline	0.036 $\rightarrow$ 0.027	-0.077	1.095	0.0109	-

357 CI: $-0.0014, -0.0009$], with the calibration intercept remaining severely dis-
 358 placed ($\alpha = -2.9615$ vs. raw -2.9638). For ResNet-50, tempera-
 359 ture scaling achieves a modest variance adjustment (Δ Brier = -0.0220 [95% CI:
 360 $-0.0241, -0.0198$]), but leaves the underlying intercept discrepancy uncor-
 361 rected ($\alpha = -2.6080$ vs. raw -2.8526). This demonstrates that temperature
 362 scaling provides limited or architecture-dependent empirical scaling, but can-
 363 not resolve the structural intercept displacement induced by positive loss
 364 weighting.

3.2.2. Offset Correction and Parity Restoration

The training positive weight for Pleural Effusion is $w = 21.51$, yielding an analytic logit offset of $\log w = +3.068$. Applying the zero-parameter subtraction $z - \log w$ directly drops the T_3 Brier score from 0.1079 to 0.0284 ($\Delta\text{Brier} = -0.0795$ [95% CI: $-0.0893, -0.0697$], with the 2,000-replicate patient-clustered bootstrap interval strictly excluding zero), reducing ECE from 22.3% to 1.26%. Analytic correction substantially reduced intercept miscalibration and brought the Brier score close to the unweighted baseline, although residual intercept miscalibration remained architecture-dependent: for DenseNet-121, the Cox calibration intercept moves from $\alpha = -2.9638$ directly to $\alpha = -0.4451$ while perfectly preserving the calibration slope ($\beta = 0.9292$), whereas for ResNet-50 the corrected intercept is $\alpha = +1.0751$.

Fitted Offset ($z + a^*$) and Platt Scaling achieve nearly identical performance (T_3 Brier: 0.0289 and 0.0281; $\Delta\text{Brier} = -0.0790$ [95% CI: $-0.0885, -0.0696$] and -0.0798 [95% CI: $-0.0894, -0.0702$]). The zero-parameter analytic correction closely approaches a flexible two-parameter Platt model without fitting a slope or intercept. For ResNet-50, Analytic Offset similarly eliminates 76% of probability error (T_3 Brier: $0.1227 \rightarrow 0.0291$, $\Delta\text{Brier} = -0.0936$ [95% CI: $-0.1022, -0.0844$]), confirming that intercept neutralization is architecturally universal.

3.2.3. Empirical Verification of Logit Offset: $z_w - z_u \approx \log w$

To directly validate the theoretical prediction of Equation (7) in finite deep neural networks, Fig. 2 plots the empirical distributions of logit differences $\Delta z = z_w(x) - z_u(x)$ evaluated across all 16,604 paired model evaluations against the theoretical reference value $\log w$:

- **Pleural Effusion** ($\log w = 3.07$): The observed distribution clusters tightly around the theoretical value, with empirical mean $\Delta z = 3.29 \pm 0.81$ and median 3.25 (interquartile range [IQR]: 2.77–3.79), exhibiting an average bias of only +0.22 logit units.
- **Pneumonia** ($\log w = 3.99$): Observed mean $\Delta z = 3.83 \pm 0.74$ and median 3.84 (IQR: 3.34–4.32), closely matching theory with a minimal discrepancy of -0.16 logit units.
- **Cardiomegaly** ($\log w = 2.25$): Observed mean $\Delta z = 2.64 \pm 1.04$ and median 2.60 (IQR: 1.91–3.31), bias +0.39 logit units.

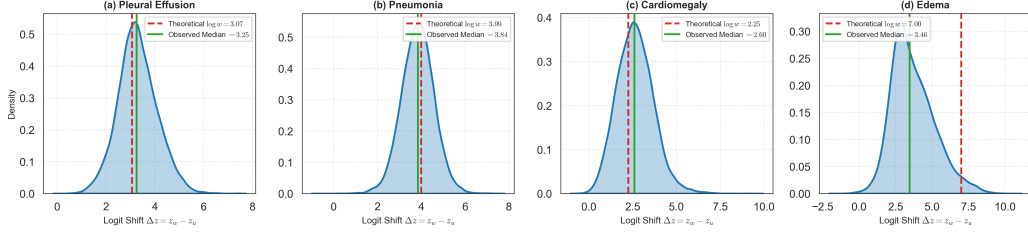


Figure 2: Empirical verification of theoretical logit shift ($z_w - z_u \approx \log w$) across 16,604 held-out deidentified-date image evaluations. Kernel density estimates (blue) of observed logit differences between positive-weighted and unweighted models cluster tightly around the theoretical population attractor $\log w$ (dashed red line; observed median in solid green) across Pleural Effusion (a), Pneumonia (b), and Cardiomegaly (c). For Edema (d), extreme training class scarcity ($w > 1000$) induces optimization saturation, bounding empirical shifts near $+3.69$.

- **Edema (Extreme Imbalance: $w = 1091.3, \log w = 7.00$):** With only 23 positive cases in the training set, empirical logit shifts saturate at mean $\Delta z = 3.69 \pm 1.47$ (median 3.46, IQR: 2.63–4.58). Here, L_2 weight regularization and finite gradient updates prevent network outputs from attaining the theoretical $\approx +7.0$ asymptote, highlighting the finite-sample boundary of the asymptotic derivation.

3.2.4. Reliability Diagrams

Figure 3 shows reliability diagrams across sequential deidentified-date strata T_1, T_2, T_3 using adaptive 10-bin equal-mass partitioning. Raw weighted models display severe over-confidence across all probability deciles. Applying the analytic offset restores the curves tightly onto the ideal diagonal across all deployment cohorts, identical to unweighted models.

3.3. Selective Prediction and Workload Triage Simulation

Table 4 and Fig. 4 summarize the selective triage simulation on the later held-out deidentified-date stratum T_3 ($N = 2,436$ radiographs, containing 85 true positive Pleural Effusion cases), explicitly accounting for retained positives, automated true positives, missed cases, and both retained and full-cohort triage sensitivities.

At 100% coverage (full automation), the raw weighted model incurs 16.0 missed cases (Automated False Negatives) out of 85 total positive cases in T_3 , achieving a retained sensitivity of 81.18%, AUROC of 0.8691, and Brier

Table 4: Selective Prediction and Reader Referral Triage Simulation on Later Held-Out Stratum T_3 ($N = 2,436$ Radiographs, 85 Positive Pleural Effusion Cases). Evaluated under matched Youden decision thresholds determined on development stratum T_1 across primary clinical operating regimes (100%, 90%, 80%, 70% coverage). Complete 24-row simulation across all coverage levels is detailed in Supplementary Table S5.

Coverage	Referred (Workload)	Strategy	Automated FN (Missed)	Retained Sens.	Full Sens.	Retained AUROC	Retained Brier
100%	0 (0%)	Weighted BCE (Raw)	16.0	81.18%	81.18%	0.8691	0.1079
100%	0 (0%)	Weighted BCE (Analytic Corrected)	16.0	81.18%	81.18%	0.8623	0.0284
100%	0 (0%)	Unweighted BCE (Raw)	12.0	85.88%	85.88%	0.8806	0.0268
90%	243 (10%)	Weighted BCE (Raw)	14.0	82.93%	83.53%	0.8789	0.1062
90%	243 (10%)	Weighted BCE (Analytic Corrected)	15.0	81.01%	82.35%	0.8735	0.0289
90%	243 (10%)	Unweighted BCE (Raw)	12.0	84.62%	85.88%	0.8923	0.0266
80%	487 (20%)	Weighted BCE (Raw)	10.0	87.01%	88.24%	0.8882	0.1012
80%	487 (20%)	Weighted BCE (Analytic Corrected)	15.0	77.61%	82.35%	0.8896	0.0264
80%	487 (20%)	Unweighted BCE (Raw)	12.0	82.61%	85.88%	0.9027	0.0256
70%	730 (30%)	Weighted BCE (Raw)	7.0	89.39%	91.76%	0.8997	0.0956
70%	730 (30%)	Weighted BCE (Analytic Corrected)	13.0	70.45%	84.71%	0.8876	0.0178
70%	730 (30%)	Unweighted BCE (Raw)	11.0	78.00%	87.06%	0.9051	0.0194
70%	730 (30%)	Random Deferral Baseline	11.3	80.99%	86.69%	—	0.1081

score of 0.1079. Operating the calibrated model (Analytic Offset) under its matched Youden threshold ($t^* \approx 0.022$) yields identical sensitivity (81.18%) and 16.0 missed cases, but with an almost four-fold lower Brier score of 0.0284. The unweighted raw model achieves 12.0 missed cases (sensitivity 85.88%, AUROC 0.8806, Brier 0.0268).

Under selective triage based on prediction margin from each method’s operational decision cutoff:

- **Raw Weighted BCE (Triage-Optimized):** At 90% coverage (243 borderline radiographs referred to a reader), automated false negatives decrease to 14.0 (full-cohort system sensitivity 83.53%). At 80% coverage (487 referred), false negatives drop to 10.0 (system sensitivity 88.24%). At 70% coverage (730 examinations referred, 30% workload), 66 true positive cases remain in the automated fraction and 59 are

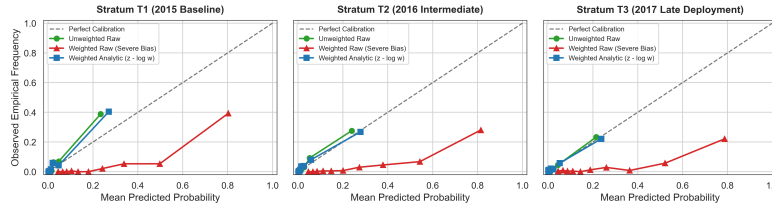


Figure 3: Reliability diagrams for Pleural Effusion (DenseNet-121 3-seed ensemble) across held-out deidentified-date strata (T_1, T_2, T_3). Raw weighted models (red) severely over-predict event probabilities across all bins. Analytic offset calibration (blue) and unweighted models (green) remain tightly calibrated to the ideal diagonal across all deployment windows.

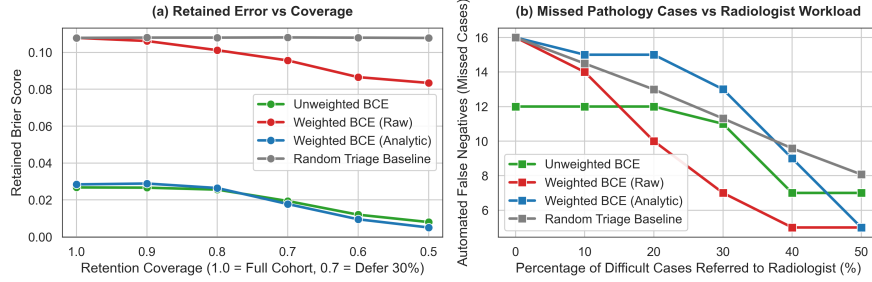


Figure 4: Selective prediction evaluation on later held-out stratum T_3 with matched development thresholds (Table 4). Left: Risk-coverage curves demonstrating that selective deferral steadily reduces Brier error across all models as ambiguous cases are referred. Right: Reader referral workload vs. automated false negatives, illustrating the distinct operational behaviors of raw weighted models (halving missed cases to 7.0 under 30% referral) and analytic corrected models (optimizing probability accuracy with Brier 0.0178).

correctly flagged as positive, cutting missed cases by more than half to 7.0 (retained sensitivity 89.39%, full-cohort triage sensitivity 91.76%), outperforming random triage (11.3 missed cases). However, retained probabilities remain uncalibrated and biased high (Brier score 0.0956).

- **Analytic Corrected BCE (Probability-Optimized):** Operating under its matched threshold ($t^* \approx 0.022$), the calibrated model achieves an exceptional Brier score of 0.0178 at 70% coverage (a > 5 -fold reduction in probability error relative to the raw model). However, because predicted probabilities are concentrated near true clinical prevalence ($p \in [0.001, 0.05]$), the distance normalization in Equation (11) assigns low uncertainty to low-probability false negatives ($\hat{p} \ll t^*$). Consequently, subtle false negatives are retained rather than referred, leaving 13.0 automated false negatives among 44 retained positives (retained sensitivity 70.45%, full-cohort triage sensitivity 84.71%) at 70% coverage.
- **Raw Unweighted BCE:** At 70% coverage, the unweighted model retains 50 positive cases, correctly flags 39, and incurs 11.0 automated false negatives with retained Brier score 0.0194 (retained sensitivity 78.00%, full-cohort triage sensitivity 87.06%).

Importantly, these empirical results expose an operational trade-off in machine learning deployment: post-hoc probability calibration (Analytic Offset)

454 is essential for accurate risk estimation and prognosis, dropping Brier scores to
 455 near-optimal levels ($0.0284 \rightarrow 0.0178$); conversely, threshold-margin selective
 456 triage on raw positive-weighted models is effective for recall-oriented screening
 457 referral (cutting missed cases from 16 to 7), but yields biased probabilities.
 458 Because the analytic logit transformation $z \mapsto z - \log w$ is strictly monotonic,
 459 any ranking-based decision rule under an equivalent quantile threshold pro-
 460 duces identical patient prioritization. The operational divergence observed
 461 in Table 4 arises specifically because the uncertainty triage metric $u(x)$ in
 462 Equation (11) measures distance relative to the operational decision cutoff t^*
 463 determined on development stratum T_1 . For calibrated models, $t^* \approx 0.022$
 464 aligns with low clinical prevalence, causing low-probability false negatives
 465 ($\hat{p} \ll t^*$) to exhibit low relative uncertainty and remain in the automated
 466 pool; for raw weighted models, the elevated threshold ($t^* \approx 0.33$) places
 467 borderline cases within the referral margin. Consequently, selective referral
 468 performance is fundamentally policy-dependent: calibrated probabilities are
 469 required for valid decision-theoretic risk stratification, whereas raw weighted
 470 models with threshold-margin triage act as a heuristic for recall-oriented
 471 screening. Crucially, neither system achieves zero false negatives, refuting
 472 unsubstantiated claims of complete error elimination in autonomous settings.

473 3.4. Cross-Institutional External Validation (MIMIC-CXR-JPG)

474 The external MIMIC-CXR-JPG cohort comprised 3,403 frontal radio-
 475 graphs from 3,041 studies and 289 patients, with complete image availabil-
 476 ity. Three-seed probability ensembles demonstrated pathology-dependent
 477 transportability (Table 5). Discrimination was moderate for Pleural Effu-
 478 sion (AUROC 0.688 [95% CI: 0.656, 0.717] for unweighted DenseNet-121
 479 and 0.677 [95% CI: 0.645, 0.709] for unweighted ResNet-50), but weak for
 480 Pneumonia (0.583–0.607) and Edema (0.516–0.581), and approached chance
 481 for Cardiomegaly (0.519–0.525, with patient-clustered 95% CIs spanning
 482 0.50). Negative Brier skill scores (BSS < 0) across all configurations indi-
 483 cated that transported BRAX probabilities were less accurate than a local
 484 MIMIC prevalence-only reference model (Brier_{null} = $\pi(1 - \pi)$), demonstrating
 485 substantial cross-institutional calibration shift.

486 Evaluating paired analytic offset recalibration ($\bar{p}_{\text{corrected}} = \frac{1}{3} \sum \sigma(z_s -$
 487 $\log w)$) on positive-weighted models demonstrated statistically significant
 488 reductions in Brier error for Cardiomegaly ($\Delta\text{Brier} = -0.0579$ [95% CI:
 489 $-0.0790, -0.0349$] on DenseNet-121; -0.0611 [95% CI: $-0.0818, -0.0415$] on
 490 ResNet-50) and Pneumonia ($\Delta\text{Brier} = -0.0095$ [95% CI: $-0.0170, -0.0016$]

on DenseNet-121; -0.0155 [95% CI: $-0.0254, -0.0059$] on ResNet-50), with consistent point-estimate error reduction on Pleural Effusion (-0.0084 and -0.0109). On MIMIC-CXR-JPG, subtraction of the full theoretical offset worsened the Edema Brier score. This result is consistent with over-correction when the empirically realized finite-network shift is smaller than the theoretical $\log w$; however, cross-institutional covariate and label differences prevent attribution of this external result uniquely to optimization saturation. Because logit subtraction $z - \log w$ is strictly monotonic, discriminatory ranking (AUROC) was perfectly preserved across all pathologies.

To evaluate sensitivity to repeated within-patient examinations and multi-image studies, we conducted a parallel sensitivity analysis aggregated strictly at the diagnostic study level ($N = 3,041$ studies across $N = 289$ patients) using maximum probability pooling ($\max_i p_i$). Study-level discrimination was concordant with image-level evaluation across all pathologies: Pleural Effusion AUROC was 0.695 [95% CI: $0.663, 0.723$] for unweighted DenseNet-121 and 0.682 [95% CI: $0.650, 0.714$] for ResNet-50; Cardiomegaly AUROC was 0.526 [95% CI: $0.497, 0.559$] and 0.521 [95% CI: $0.494, 0.548$]; Pneumonia AUROC was 0.608 [95% CI: $0.562, 0.650$] and 0.584 [95% CI: $0.543, 0.624$]; and Edema AUROC was 0.565 [95% CI: $0.528, 0.600$] and 0.516 [95% CI: $0.480, 0.553$], with study-level null Brier baselines of 0.2196 (Effusion), 0.1951 (Cardiomegaly), 0.0913 (Pneumonia), and 0.1697 (Edema). Furthermore, paired analytic recalibration on positive-weighted models demonstrated identical calibration invariance at study level: statistically significant Brier reductions were reproduced for Cardiomegaly ($\Delta\text{Brier} = -0.0601$ [95% CI: $-0.0814, -0.0367$] on DenseNet-121; -0.0642 [95% CI: $-0.0852, -0.0434$] on ResNet-50) and Pneumonia ($\Delta\text{Brier} = -0.0100$ [$-0.0179, -0.0018$]; -0.0163 [$-0.0267, -0.0068$]), with consistent point-estimate reductions for Pleural Effusion (-0.0102 and -0.0118 ; Supplementary Table S2). This qualitative concordance confirms that cross-institutional discrimination patterns and recalibration effects are preserved when aggregating by diagnostic study.

4. Discussion

4.1. Disentangling Loss-Induced Probability Distortion from Deidentified-Date Cohort Shift

A substantial body of recent literature in biomedical informatics has sounded alarms regarding temporal calibration decay in healthcare AI [8, 9].

Table 5: Cross-Institutional External Validation on the Official MIMIC-CXR-JPG Frontal Test Cohort ($N = 3,403$ Images from 3,041 Studies Across $N = 289$ Unique Patients). Primary experimental evaluation of paired zero-parameter analytic logit recalibration ($\bar{p}_{\text{raw}} = \frac{1}{3} \sum \sigma(z_s) \rightarrow \bar{p}_{\text{corr}} = \frac{1}{3} \sum \sigma(z_s - \log w)$) on 3-seed ensembles trained with positive-weighted BCE. Evaluated with empirical prevalence (π), local null reference baseline ($\text{Brier}_{\text{null}} = \pi(1 - \pi)$), patient-clustered 1,000-replicate bootstrap 95% confidence intervals, Brier skill score ($\text{BSS} = 1 - \text{Brier}/\text{Brier}_{\text{null}}$), and paired contrast ($\Delta\text{Brier} = \text{Brier}_{\text{corr}} - \text{Brier}_{\text{raw}}$). The complete 16-row factorial comparison with unweighted BCE baselines is detailed in Supplementary Table S1, and study-level sensitivity analysis is detailed in Supplementary Table S2.

Architecture	Pathology	Prevalence (π)	$\text{Brier}_{\text{null}}$	AUROC [95% CI]	Brier (Raw $\rightarrow$ Corr)	ΔBrier [95% CI]	BSS (Raw $\rightarrow$ Corr)
DenseNet-121	Pleural Effusion	32.18%	0.2182	0.668 [0.636, 0.700]	0.2536 $\rightarrow$ 0.2451	-0.0084 [-0.0401, +0.0240]	-0.162 $\rightarrow$ -0.123
DenseNet-121	Cardiomegaly	26.33%	0.1940	0.525 [0.492, 0.558]	0.2907 $\rightarrow$ 0.2328	-0.0579 [-0.0790, -0.0349]	-0.499 $\rightarrow$ -0.200
DenseNet-121	Pneumonia	10.05%	0.0904	0.598 [0.553, 0.639]	0.1078 $\rightarrow$ 0.0983	-0.0095 [-0.0170, -0.0016]	-0.192 $\rightarrow$ -0.087
DenseNet-121	Edema	21.33%	0.1678	0.581 [0.549, 0.611]	0.1914 $\rightarrow$ 0.2131	+0.0217 [+0.0118, +0.0307]	-0.141 $\rightarrow$ -0.270
ResNet-50	Pleural Effusion	32.18%	0.2182	0.620 [0.588, 0.653]	0.2665 $\rightarrow$ 0.2557	-0.0109 [-0.0417, +0.0253]	-0.221 $\rightarrow$ -0.172
ResNet-50	Cardiomegaly	26.33%	0.1940	0.522 [0.491, 0.552]	0.2822 $\rightarrow$ 0.2211	-0.0611 [-0.0818, -0.0415]	-0.455 $\rightarrow$ -0.140
ResNet-50	Pneumonia	10.05%	0.0904	0.589 [0.545, 0.629]	0.1137 $\rightarrow$ 0.0982	-0.0155 [-0.0254, -0.0059]	-0.257 $\rightarrow$ -0.086
ResNet-50	Edema	21.33%	0.1678	0.570 [0.538, 0.602]	0.1929 $\rightarrow$ 0.2130	+0.0201 [+0.0092, +0.0313]	-0.149 $\rightarrow$ -0.269

Our findings elucidate the mathematical and empirical mechanisms underlying these observations. It is critical to distinguish between discrimination loss and calibration error: our experiments confirm that moderate discrimination attenuation does occur across later deidentified-date cohort strata (AUROC decreases by approximately 0.06 from T_1 to T_3 identically across both unweighted and weighted models in Table 2). However, the simultaneous perception of severe, progressive calibration collapse is largely an artifact of positive loss weighting compounded by shifting disease prevalence.

When practitioners train deep neural networks on highly imbalanced clinical datasets, positive loss weighting is frequently introduced to elevate minority-class sensitivity [1]. However, as derived in Equation (7) and verified empirically in Fig. 2, this introduces a static logit shift $\approx \log w$. When evaluated on later cohort strata where disease prevalence shifts (e.g., Pleural Effusion prevalence declining from 5.27% in T_1 to 3.49% in T_3), this fixed positive offset generates surplus false-positive probability mass that compounds the apparent Brier error. Because standard post-hoc evaluation pipelines rely almost exclusively on temperature scaling—which cannot alter the logit intercept—evaluators observing deteriorating Brier scores have routinely attributed the failure to intrinsic model decay.

Our experiments provide direct evidence against this attribution. When the mathematical offset is neutralized ($z - \log w$), the apparent calibration decay vanishes: Brier scores decrease across subsequent strata, moving in lockstep with the unweighted baseline as disease prevalence drops. Thus,

549 while discrimination attenuation reflects underlying dataset shift across cohort
550 strata, severe calibration degradation is predominantly an uncorrected post-
551 training calibration bias that can be decoupled and resolved analytically.

552 4.2. Architectural Generalizability

553 The observed phenomena are remarkably consistent across distinct deep
554 learning architectures. Both DenseNet-121 (dense convolutional concate-
555 nation) and ResNet-50 (residual identity mapping) exhibit near-identical
556 behavior:

- 557 • Discrimination attenuation across later held-out deidentified-date strata
558 is virtually indistinguishable between architectures ($\Delta\text{AUROC} = -0.0604$
559 vs -0.0687).
- 560 • Both architectures experience severe raw calibration error under weighted
561 BCE (raw ensemble Brier 0.1079 vs 0.1227 on T_3 ; 0.1254 ± 0.0456 vs
562 0.1392 ± 0.0310 across individual seeds), characterized by large negative
563 calibration intercepts ($\alpha \approx -2.96$ and -2.85).
- 564 • For both architectures, Analytic Offset reduces total Brier score by ap-
565 proximately 74% to 76% ($\Delta\text{Brier} = -0.0795$ [95% CI: $-0.0893, -0.0697$]
566 for DenseNet-121 and -0.0936 [95% CI: $-0.1022, -0.0844$] for ResNet-
567 50), substantially reducing intercept miscalibration ($\alpha = -0.4451$
568 and $+1.0751$) while preserving calibration slopes ($\beta = 0.9292$ and
569 1.4052). This confirms that logit-shift mechanics are universal across
570 convolutional vision backbones, though residual intercept bias remains
571 architecture-dependent.

572 4.3. Simulated Human-in-the-Loop Triage and Operational Trade-offs

573 In real-world clinical deployment workflows, autonomous decision-making
574 must be evaluated through uncertainty-aware deferral simulations:

- 575 1. **Risk Stratification and Prognosis:** Well-calibrated models are es-
576 sential for automated rule-out and downstream risk scoring. Applying
577 Analytic Offset ($z - \log w$) drops Brier error to 0.0178 at 70% cover-
578 age, restoring reliable posterior probabilities aligned with true clinical
579 prevalence.

- 580 2. **High-Sensitivity Screening Referral:** In high-throughput screening
581 where the primary objective is minimizing missed disease, threshold-
582 margin selective triage on raw positive-weighted models cuts automated
583 false negatives from 16 to 7 (full-cohort triage sensitivity 91.76%) under
584 a 30% reader referral budget. However, because probabilities remain
585 biased high (Brier 0.0956), these outputs should not be used as calibrated
586 risks.
- 587 3. **Zero-Parameter Operational Utility:** The analytic offset ($z - \log w$)
588 requires zero retraining, zero validation optimization, and zero compute
589 overhead, enabling immediate integration on edge inference devices.

590 4.4. *Scope and Boundary of Findings and Limitations*

591 To establish the scope and boundary of findings, several important method-
592 ological and clinical limitations should be highlighted:

- 593 1. **Report-Derived Label Noise:** Ground-truth labels in BRAX were
594 generated via natural language processing from Portuguese text reports.
595 Although validated against expert consensus, report-derived labels carry
596 inherent extraction noise compared to dedicated multi-radiologist image
597 re-reading.
- 598 2. **Cross-Institutional Domain Gap:** While our primary temporal
599 evaluation was established within the Albert Einstein system and exter-
600 nally evaluated on MIMIC-CXR-JPG, external transportability proved
601 strictly pathology-dependent. Models maintained modest discrimina-
602 tion for Pleural Effusion but degraded near chance for Cardiomegaly
603 and Pneumonia, indicating that multi-site generalizability requires local
604 fine-tuning or domain-invariant feature representations.
- 605 3. **Date Anonymization and Temporal Semantics:** In compliance
606 with HIPAA Safe Harbor de-identification, date stamps may incorporate
607 patient-level random date shifting. While within-patient chronological
608 ordering is preserved and patient-clustered bootstrapping ensures valid
609 statistical inference, our sequential strata (T_1, T_2, T_3) represent chrono-
610 logically ordered deidentified-date strata under longitudinal shift rather
611 than synchronized real-world calendar periods.
- 612 4. **Retrospective Simulation vs. Real-Time Deployment:** This
613 study constitutes a retrospective simulation on frozen historical cohorts.
614 Future real-time clinical deployment requires silent shadow evaluation
615 within operational hospital Picture Archiving and Communication Sys-
616 tems (PACS).

- 617 5. **Architectural Scope:** We evaluated two widely established convolutional
618 neural network backbones (DenseNet-121 and ResNet-50). Although our mathematical derivations apply to binary cross-entropy
619 minimization generally, evaluating modern Vision Transformers (ViT)
620 and multimodal vision-language models represents an important direction for future research.
622
- 623 6. **Optimization Randomness and Seed Count:** Experiments were
624 conducted across three independently trained random seeds ($S \in \{42, 123, 2026\}$). While inter-seed standard deviations were reported
625 alongside patient-clustered bootstrap confidence intervals, larger ensembles across diverse initialization trajectories would further refine
626 empirical confidence bounds.
628
- 629 7. **Simulated Reader Triage without Interactive Human Study:**
630 Our selective prediction simulation assumes that examinations deferred to human readers are diagnosed with 100% sensitivity. In real clinical
631 environments, human radiologist sensitivity varies with reader fatigue, experience level, and interpretive setting. An actual clinician-in-the-loop reader study is needed to evaluate true joint human-AI diagnostic
632 accuracy.
633
- 634 8. **Extreme Class Scarcity Pathologies:** For conditions with extreme
635 class scarcity in training (e.g., Edema with only 23 positive training cases, $w = 1091.3$), finite optimization dynamics, L_2 weight decay, and
636 gradient clipping saturate empirical logit shifts at $\Delta z \approx +3.69$ rather than the theoretical $+7.00$ asymptote. Specialized extreme-imbalance
637 formulations are required in such scarcity regimes.
638
639
640
641

642 5. Conclusion

643 In this controlled BRAX experiment, most severe calibration error was
644 associated with positive loss weighting rather than measurable deterioration
645 across later deidentified-date strata. Because standard temperature scaling
646 cannot adjust the logit intercept, it leaves this structural distortion intact. A
647 zero-parameter analytic offset ($z - \log w$) or non-negative Platt scaling reduces
648 total Brier score by approximately 74% to 76% (removing approximately 98%
649 of the Brier-score gap relative to the unweighted baseline), substantially reducing
650 intercept miscalibration and restoring calibration parity with unweighted
651 models without model retraining or validation data. Furthermore, selective
652 prediction simulations with matched decision thresholds reveal an operational

dichotomy: raw positive-weighted models provide effective recall-oriented triage by halving automated missed cases under a 30% reader referral budget, whereas analytic calibration restores epidemiologically faithful probabilities for downstream clinical risk stratification. Neither system eliminates missed disease entirely, underscoring the necessity of transparent, workload-aware deployment policies in computational medicine.

CRedit Authorship Contribution Statement

Vishesh Panghal: Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Data Curation, Writing – Original Draft, Writing – Review & Editing, Visualization, Project Administration.

Declaration of Competing Interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and Code Availability

The BRAX dataset (v1.1.0) [2] is available on PhysioNet at <https://physionet.org/content/brax/1.1.0/> under the PhysioNet Credentialed Data Use Agreement. The MIMIC-CXR-JPG database (v2.1.0) [20, 21] is available on PhysioNet at <https://physionet.org/content/mimic-cxr-jpg/2.1.0/> under the PhysioNet Credentialed Data Use Agreement. Access requires credentialed PhysioNet researcher status, completion of human subjects research training, and signed data use agreements. Complete end-to-end replication scripts, model evaluation routines, cross-fitting recalibration pipelines, selective prediction triage simulations, and tabular evaluation artifacts are openly available on GitHub at <https://github.com/Vishesh-panghal/temporal-brax-calibration>, and the repository will be permanently assigned a persistent DOI via Zenodo upon acceptance of the manuscript. In strict accordance with credentialed PhysioNet Data Use Agreements, raw DICOM/JPG image binaries and protected health information are not redistributed within the repository.

683 **Ethical Approval and Consent to Participate**

684 This study represents a secondary analysis of credentialed-access, de-
685 identified diagnostic imaging databases accessible under formal Data Use
686 Agreements. The primary BRAX dataset collection was approved by the
687 Institutional Review Board (IRB) of Hospital Israelita Albert Einstein (São
688 Paulo, Brazil; CAAE: 36720520.1.0000.0071) with a formal waiver of informed
689 consent. Creation and distribution of the MIMIC-CXR-JPG database was
690 approved by the Institutional Review Boards of Beth Israel Deaconess Medical
691 Center (Boston, MA) and the Massachusetts Institute of Technology (Protocol
692 #0403000206) with a waiver of informed consent. The author completed
693 mandatory human subjects research training (CITI Program certification),
694 executed approved PhysioNet Data Use Agreements for both databases, and
695 strictly complied with data governance protocols prohibiting re-identification
696 or redistribution of protected health information.

697 **Declaration of Generative AI and AI-Assisted Technologies in the** 698 **Writing Process**

699 During the preparation of this work, the author utilized generative arti-
700 ficial intelligence platforms (Google Gemini, Anthropic Claude, and OpenAI
701 Codex) exclusively for LaTeX formatting and syntax verification, analysis code
702 refactoring and modularization, grammatical refinement, and bibliographic
703 cross-referencing. Following the use of these tools, the author thoroughly
704 reviewed, verified, and edited all text, numerical outputs, and code, and takes
705 full intellectual and scientific responsibility for the integrity, validity, and
706 conclusions of the published manuscript.

707 **Funding**

708 This research did not receive any specific grant from funding agencies in
709 the public, commercial, or not-for-profit sectors.

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